

INTRODUCTION



Stroke is the second leading cause of death globally, with up to 80% of strokes preventable through early detection and intervention. This study applies systematic EDA and machine learning to a 5110 patient clinical dataset to identify the strongest predictions of strokes and translate findings into actionable clinical tools.

RECOMMENDATIONS & FUTURE WORK

We recommend deploying the risk score in HER systems, adding annual screening for patients over 60 with any comorbidity, & setting a glucose alert at 114 mg/dL. For future work, we will apply ML to boost prediction accuracy

RESEARCH QUESTIONS

Q1 Why is the dataset imbalanced?
A: With strokes at only 4.87%, the dataset is imbalanced — future models will use SMOTE.

Q2 Why did smoking show no significance?
A: With 30% of smoking data missing, the real relationship is likely masked; full data would likely show significance.

Q3 How reliable is the risk score?
A: It correctly identifies the high- risk group with a 18.24% stroke rate vs 4.87%

METHODOLOGY

Our analysis followed a four-phase pipeline: data cleaning, exploratory visuals, statistical tests using T-Tests, ANOVA, Chi-Square, and Quartiles, & patient segmentation.

Findings 1: Age
-The T-Tests shows $p < 0.0001$ confirming a strong effect
-Patients 60+ make up 71% of all strokes, despite being only 25% of the population

Findings 2: Hypertension, Heart disease & Glucose
-Hypertension triples stroke risk (13.25% vs. 3.97%)
-Heart disease quadruples it, both with Chi-Square $p < 0.0001$
-Highest glucose quartile patients have 6x's higher stroke rates than the lower ones

KEY FINDINGS (CHARTS & TABLES

